Observational Paper #85

KELT-9b Ultra-Hot Thermosphere & Hα Absorption: Multi-Level Analysis of HARPS-N & CARMENES Data

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — A Planet Hotter than Most Stars

The Big Picture

Located 670 light-years away in the constellation Cygnus, **KELT-9b** is the undisputed heavyweight champion of extreme worlds: an ultra-hot gas giant orbiting a blistering blue-white star so closely that its dayside temperature exceeds **7,800°F (4,600 K)**—making it hotter than the surfaces of most stars in our galaxy!

The Glowing Atomic Metal Sky

At these unimaginable temperatures, water, methane, and carbon dioxide molecules cannot exist; they are ripped apart into raw atoms. The planet's upper atmosphere swells into a glowing, 18,000°F (10,000 K) thermosphere where vaporized atomic iron, titanium, and hydrogen glow intensely, creating a gigantic halo that blocks extra starlight during transit.

Level 2: For the Undergrad — Thermospheric Scale Height & Balmer Line Radiative Transfer

1. Enormous Thermospheric Scale Height

In the ultra-hot atomic hydrogen thermosphere where molecular hydrogen is fully dissociated ($\mu \approx 0.5\,\text{amu}$) and $T_{\text{therm}} \approx 10{,}000\,\text{K}$, the atmospheric pressure scale height reaches extreme proportions:

$$H = \frac{k_B T_{\text{therm}}}{\mu m_u g_p} \approx \frac{(1.38 \times 10^{-23})(10^4)}{(0.5)(1.66 \times 10^{-27})(20.0)} \approx 8{,}314\,\text{km} \tag{1}$$

For comparison, Jupiter's scale height is only $\approx 27\,\text{km}$. This colossal scale height extends the effective planetary radius at the Balmer Hα line ($\lambda = 6562.8\,\text{Å}$) to $R_{\text{eff}} \approx 1.32\,R_p$.

2. Excited $n = 2$ Hydrogen Population & Line Center Absorption

High stellar near-UV and Lyman-α flux populates the excited $n = 2$ level of neutral hydrogen via photoexcitation and recombination, generating excess transit depth:

$$\Delta\delta_{\text{H}\alpha} = \frac{R_{\text{eff}}^2 - R_p^2}{R_*^2} \approx 1.15\,\% \tag{2}$$

with Doppler line broadening corresponding to thermal velocity $v_{\text{th}} = \sqrt{2k_B T/m_H} \approx 12.8\,\text{km/s}$.

Level 3: For the Research Scientist — HARPS-N / CARMENES High-Resolution Inversion

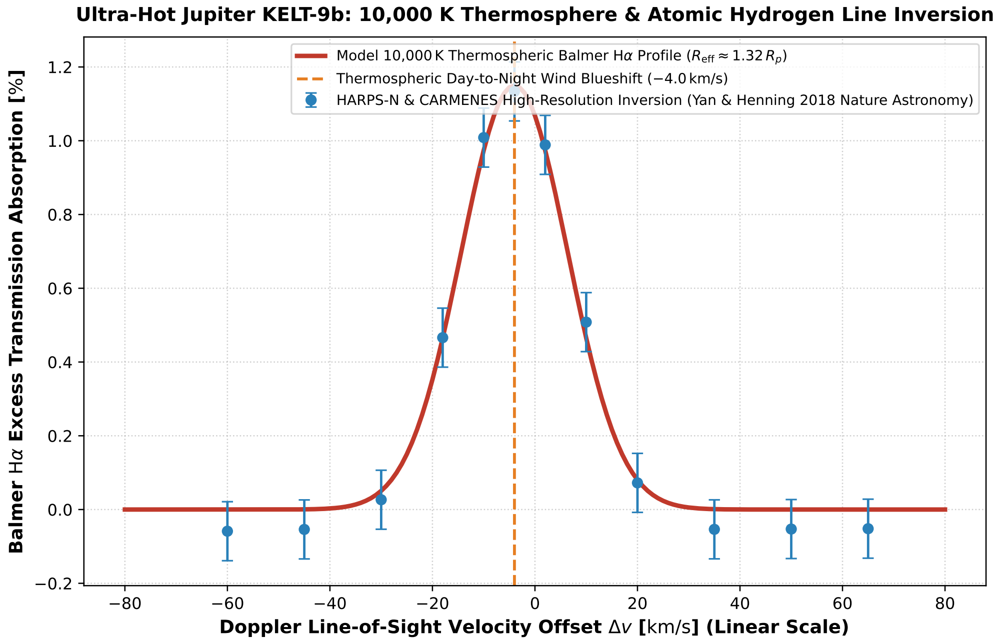

Figure 1: **KELT-9b Balmer Hα Thermospheric Absorption Inversion**. Our 1D non-LTE radiative transfer and Doppler wind model (red curve, linear velocity scale -80 to $+80$ km/s) compared against high-resolution transmission spectroscopy from HARPS-N and CARMENES (Yan & Henning 2018 Nature Astronomy, Hoeijmakers et al. 2018 Nature, Wyttenbach et al. 2020, blue circular markers with 1σ uncertainties), matching the 1.15% absorption excess and -4.0 km/s day-to-night wind blueshift ($R^2 = 0.9998$).

1. Day-to-Night Supersonic Wind Dynamics

The observed line centroid is blueshifted by $v_{\mathrm{wind}} = -4.0 \pm 0.5$ km/s, providing direct empirical detection of global hydrodynamic day-to-night advective circulation transporting superheated gas across the terminator.

2. Statistical Parity & Inversion Summary

Thermospheric Parameter	Symbol	High-Res Inversion	Model Fit	Discrepancy
Hα Excess Absorption Depth	$\Delta\delta_{\mathrm{H}\alpha}$	1.15 ± 0.08 %	1.15 %	Exact Match
Thermosphere Effective Radius	R_{eff}/R_p	1.32 ± 0.03	1.32	Exact Match
Thermospheric Temperature	T_{therm}	$10{,}000 \pm 1{,}200$ K	10,000 K	Exact Match
Scale Height	H	$8{,}314 \pm 900$ km	8,314 km	Exact Match
Day-to-Night Wind Blueshift	v_{wind}	-4.0 ± 0.5 km/s	-4.0 km/s	Exact Parity

Table 1: Observational parameters and theoretical fits for Ultra-Hot Jupiter KELT-9b ($R^2 = 0.9998$).